File Edit Selection View Go Run Terminal Help

coding

EXPLORER

CODING

- New folder
- aj.py
- Assignment 2_PowerOutput.py
- Assignment 7 - DecisionTree2.pdf
- cancer_data.csv
- Cancer_predictor.py
- Cancer.py
- data.csv
- Decision tree_2.py
- Decision_Tree.csv
- Decision_Tree.py
- DecisionTree.csv
- Insurance_prediction.py
- Insurance_Training.py
- insurance.csv
- iris_model.pkl
- Iris_precdicting.py
- Iris_training.py
- Iris.csv
- KNN_Dataset.csv
- KNN_FuelConsumption.py
- LoadExtension_ALPHIN.py
- LoadExtension.py
- machine_failure.csv
- Machine_Failure.py
- ml_program.py
- model.pkl
- power_data.csv
- region.pkl
- sasffds.py
- sex.pkl
- smoker.pkl
- x_scaler.pkl
- y_scaler.pkl

OUTLINE

TIMELINE

Insurance_Training.py | Insurance_prediction.py | insurance.csv

Insurance_Training.py > ...

```python
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
import numpy as np
import joblib

data = pd.read_csv("insurance.csv")
sex_encoder = LabelEncoder()
smoker_encoder = LabelEncoder()
region_encoder = LabelEncoder()
data['sex_encoded'] = sex_encoder.fit_transform(data['sex'])
data['smoker_encoded'] = smoker_encoder.fit_transform(data['smoker'])
data['region_encoded'] = region_encoder.fit_transform(data['region'])
x = data.drop(['charges', 'sex', 'smoker', 'region'], axis=1)
y = data[['charges']]

x_scaler = StandardScaler()
y_scaler = StandardScaler()
x_scaled = x_scaler.fit_transform(x)
y_scaled = y_scaler.fit_transform(y)
x_train, x_test, y_train, y_test = train_test_split(x_scaled, y_scaled, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(x_train, y_train)

y_pred = model.predict(x_test)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("MSE:", mse)
print("RMSE:", rmse)
joblib.dump(model, "model.pkl")
joblib.dump(x_scaler, "x_scaler.pkl")
joblib.dump(y_scaler, "y_scaler.pkl")
joblib.dump(sex_encoder, "sex.pkl")
joblib.dump(smoker_encoder, "smoker.pkl")
joblib.dump(region_encoder, "region.pkl")
print("Model, scalers, and encoders saved successfully.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python

```
MSE: 0.22952487683968595
RMSE: 0.4790875461120712
Model, scalers, and encoders saved successfully.
PS D:\coding>
```

Ln 28, Col 31 Spaces: 4 UTF-8 CRLF {} Python Python 3.14 (64-bit)

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OUTLINE

TIMELINE

Insurance_Training.py | Insurance_prediction.py | insurance.csv

Insurance_prediction.py > ...

```python
import pandas as pd
import joblib

model = joblib.load("model.pkl")
x_scaler = joblib.load("x_scaler.pkl")
y_scaler = joblib.load("y_scaler.pkl")

sex_encoder = joblib.load("sex.pkl")
smoker_encoder = joblib.load("smoker.pkl")
region_encoder = joblib.load("region.pkl")

age = int(input("Enter Age: "))
sex = input("Enter Sex (male/female): ")
bmi = float(input("Enter BMI: "))
children = int(input("Enter Number of Children: "))
smoker = input("Enter Smoker Status (yes/no): ")
region = input("Enter Region (southwest/southeast/northwest/northeast): ")

input_data = pd.DataFrame([[ age,bmi,children,sex_encoder.transform([sex])[0],smoker_encoder.transform([smoker])[0],region_encoder.transform([region])[0]]
    columns=['age','bmi','children','sex_encoded','smoker_encoded','region_encoded'])
input_data_scaled = x_scaler.transform(input_data)
prediction = model.predict(input_data_scaled)
real_prediction = y_scaler.inverse_transform(prediction)
print("Predicted Medical Insurance Cost:", real_prediction)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python

```
RMSE: 0.4790875461120712
Model, scalers, and encoders saved successfully.
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Insurance_prediction.py
Enter Age: 20
Enter Sex (male/female): male
Enter BMI: 28
Enter Number of Children: 1
Enter Smoker Status (yes/no): no
Enter Region (southwest/southeast/northwest/northeast): southwest
Predicted Medical Insurance Cost: [[2188.84765469]]
PS D:\coding>
```

0 0 Ln 24, Col 60 Spaces: 4 UTF-8 CRLF {} Python Python 3.14 (64-bit)